

## Track Generation Report

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Generated: 2026-05-30 13:40:58

Gate file: ref\_gates\_gates.csv

Gate count: 6

### Optimization parameters

method: L-BFGS-B

maxiter: 500

ftol: 1e-07

### Point counts

objective function points: 31

path generation points: 31

### Results

total optimization time: 14.681 s

total path length: 190.6189 m

### Per-step time estimate (one measured objective evaluation)

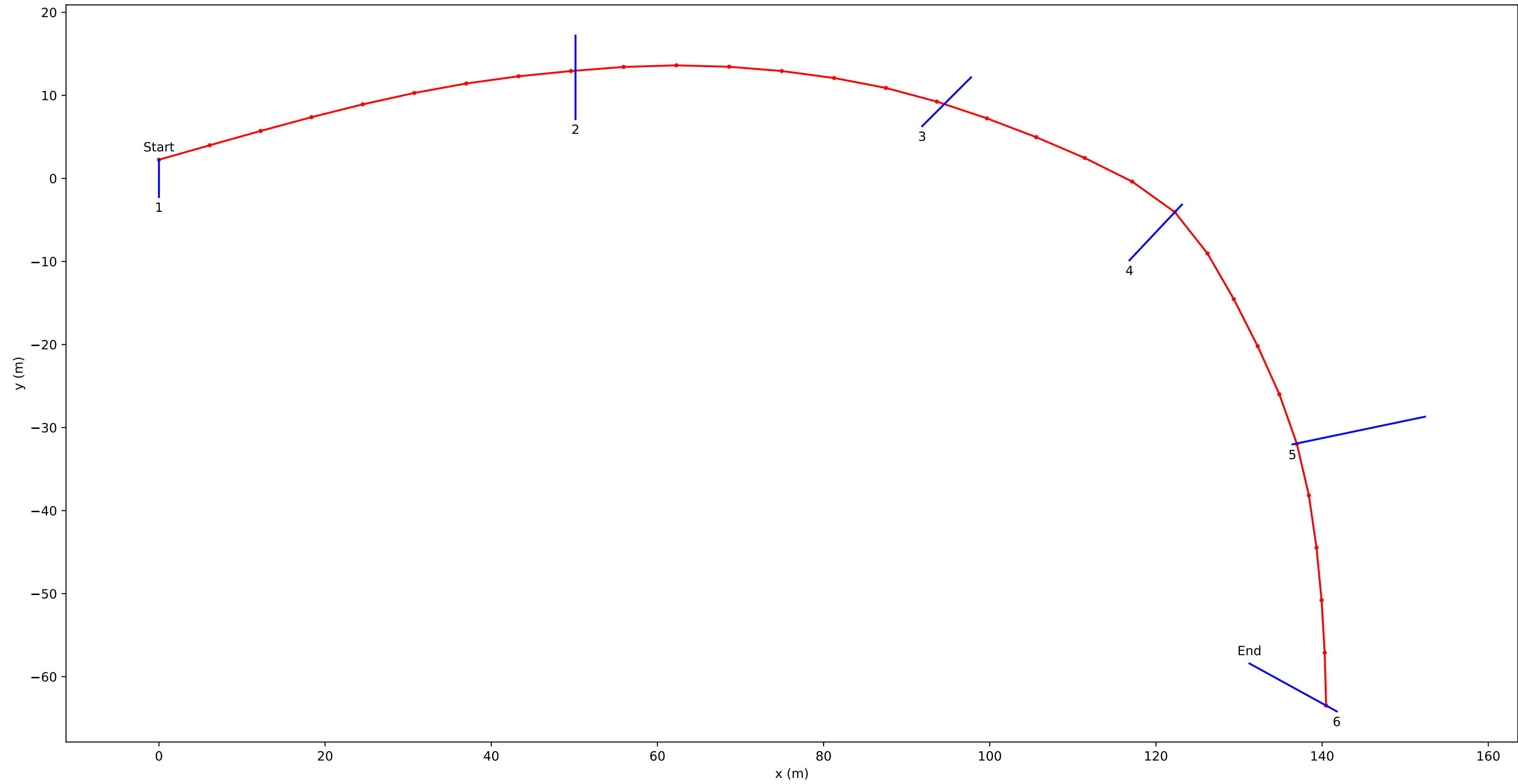
make spline from dofs: 0.89 ms

path length function: 0.60 ms

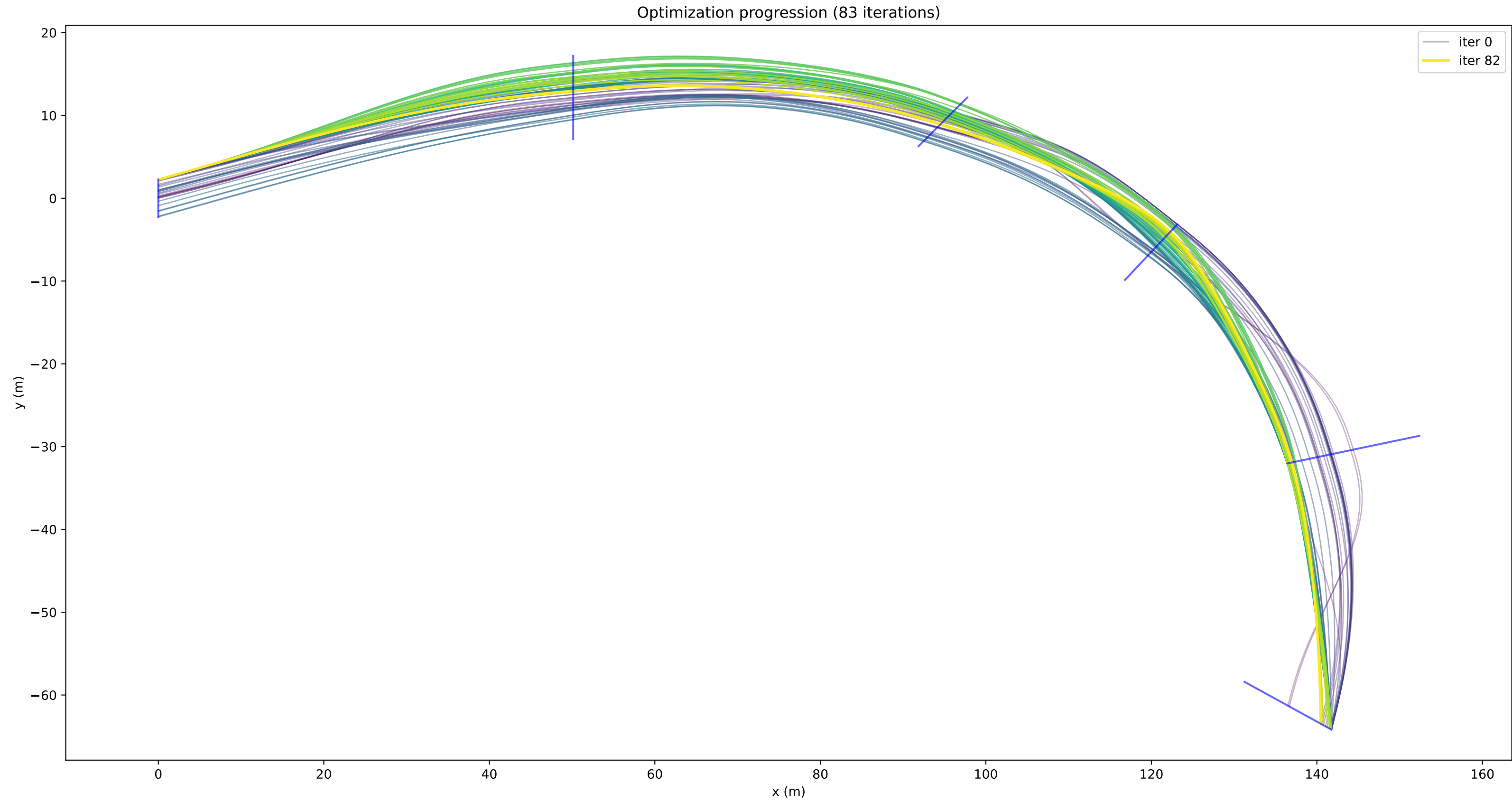
equidistant times generated: 0.51 ms

curvature objective function: 4.83 ms

Map plot: optimized path through gates

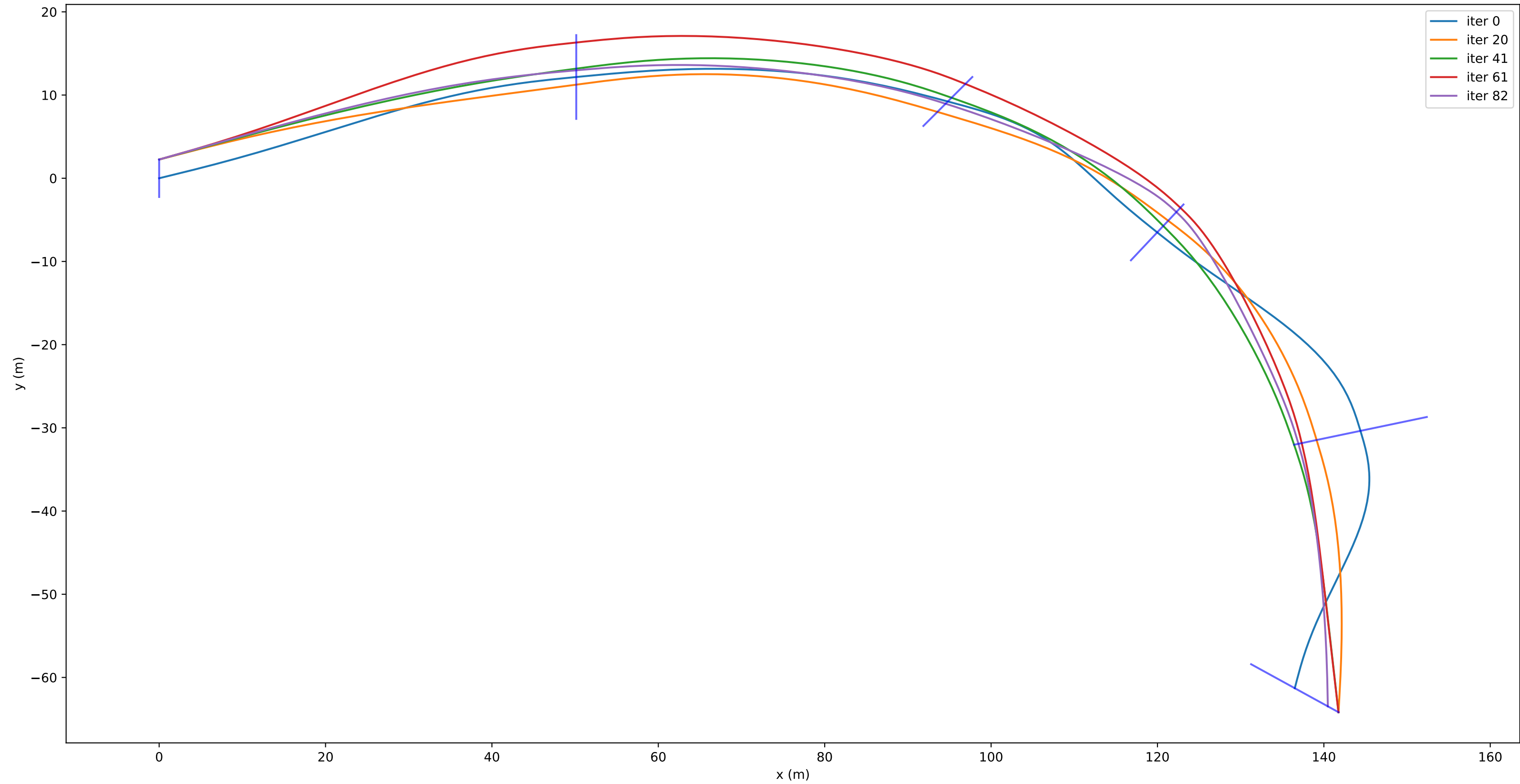


Optimization progression

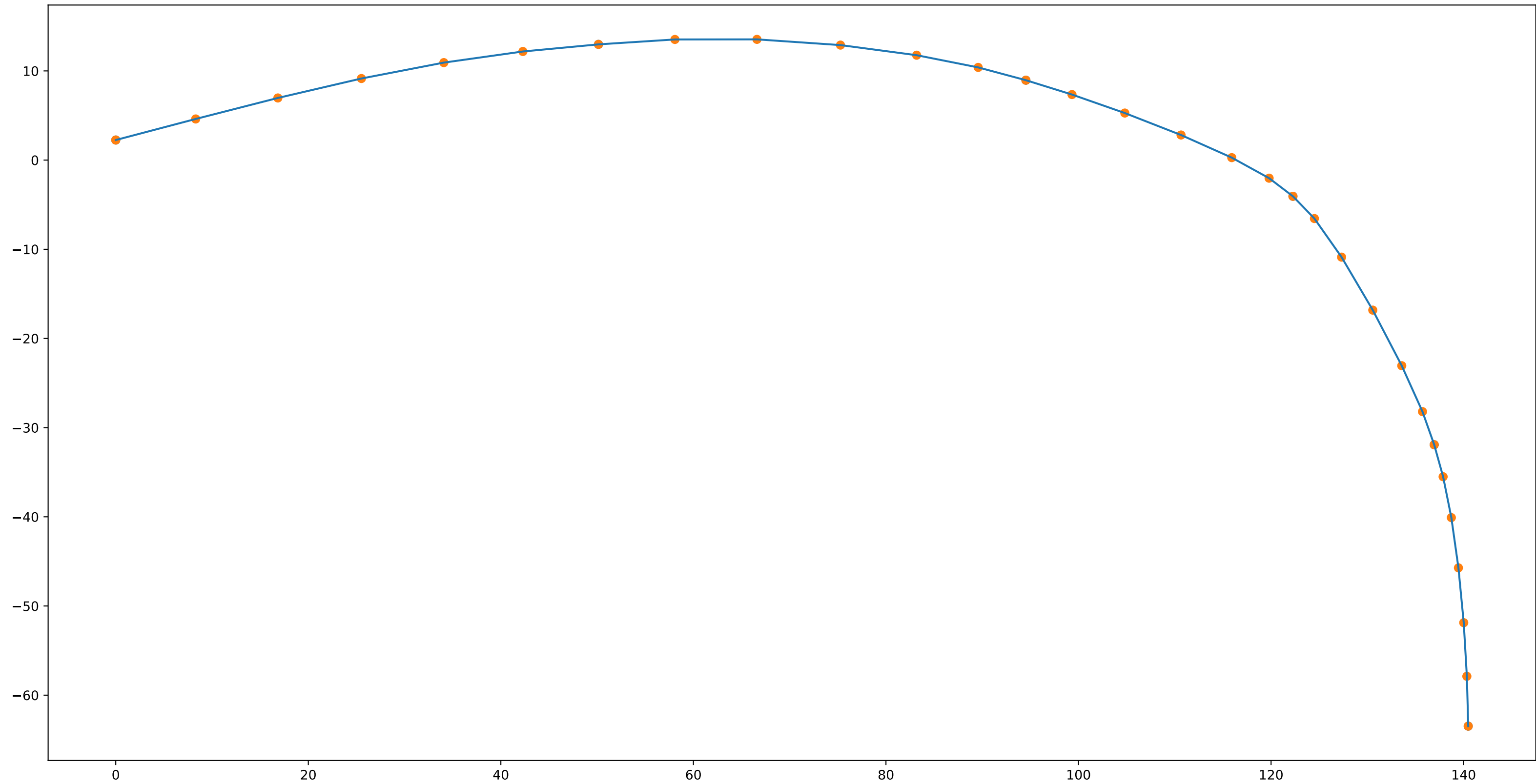


Equally spaced iterations

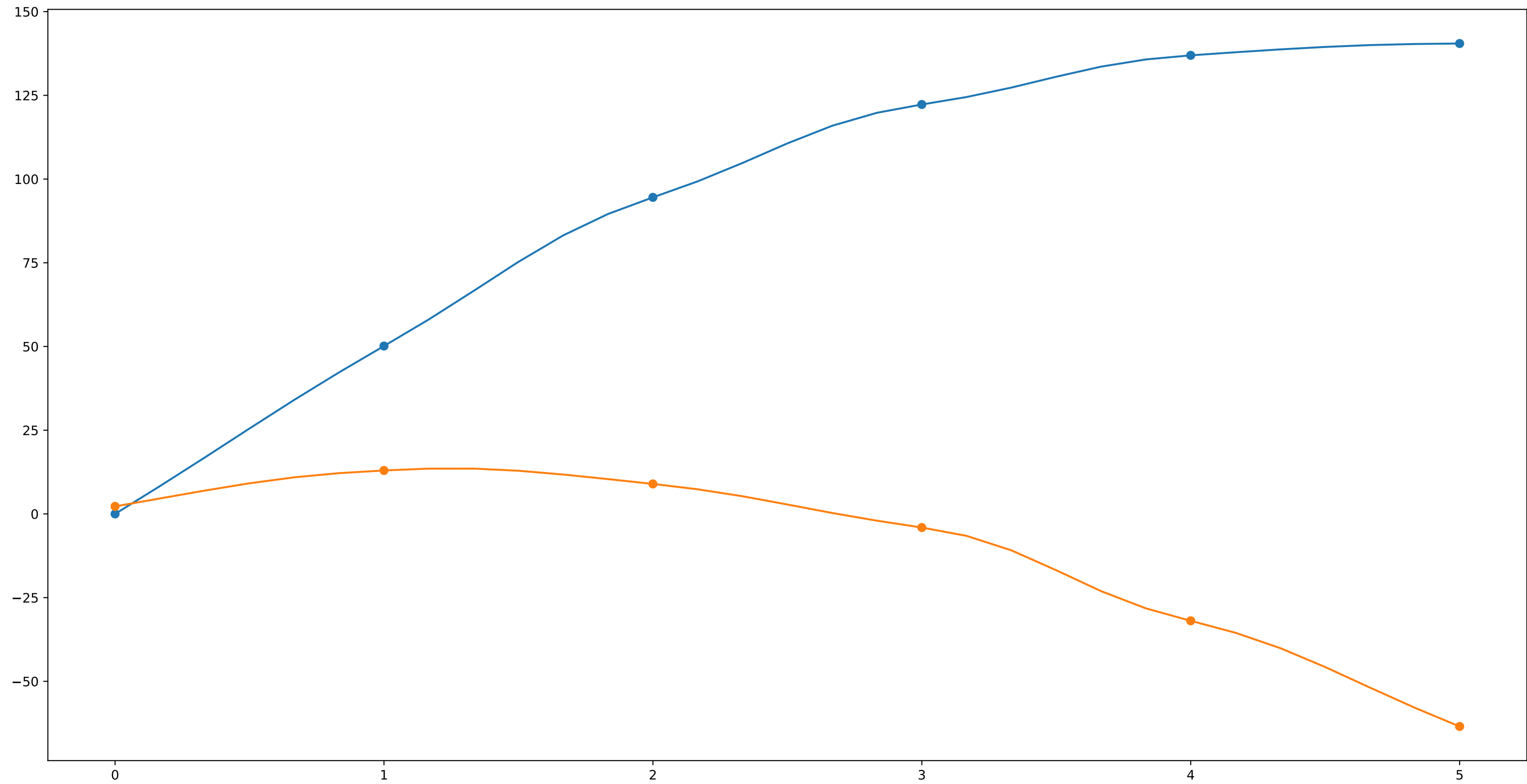
5 equally spaced iterations



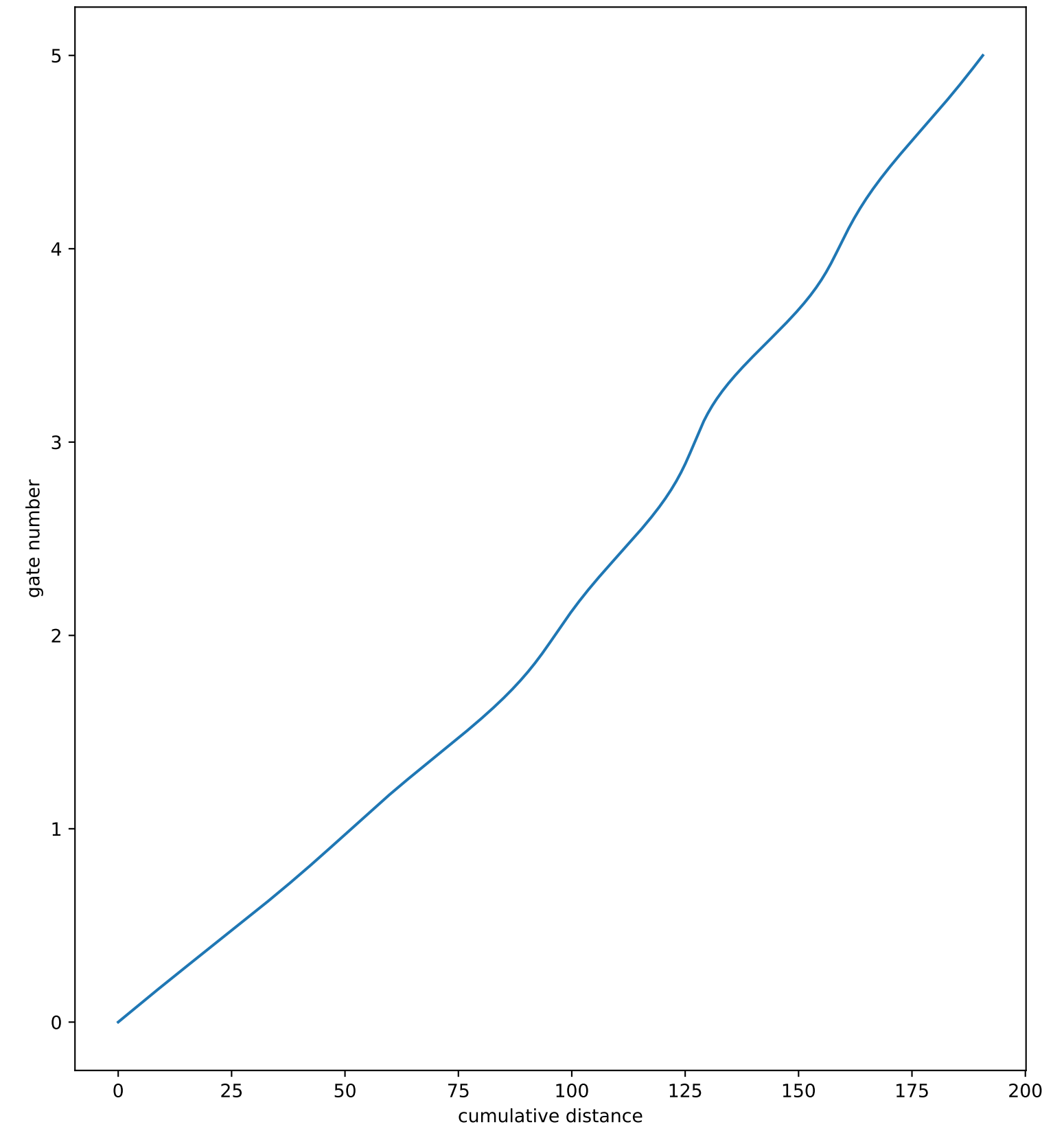
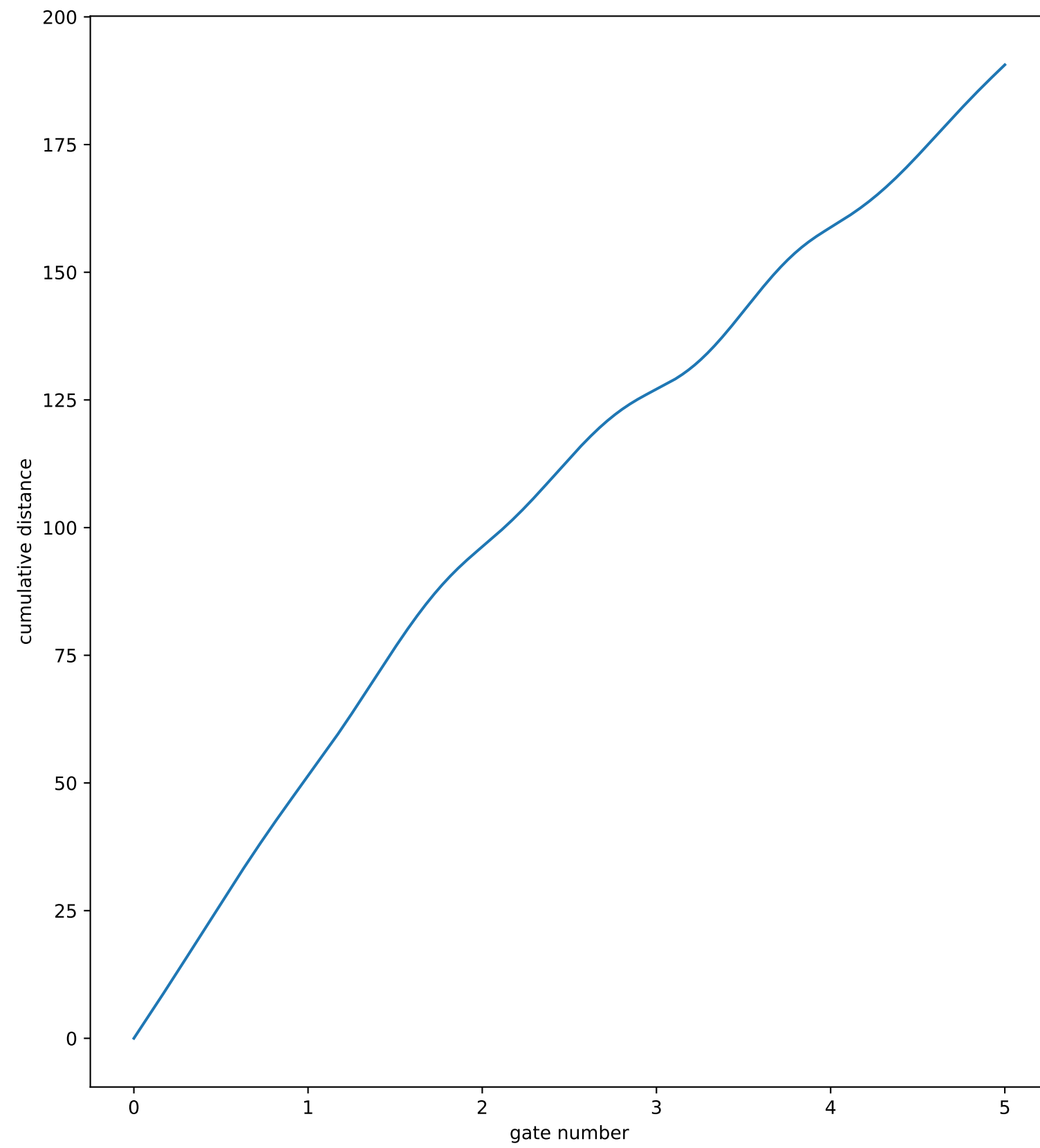
Spline debug: path and gate cross points



Spline debug: x(t) and y(t)



Path debug: cumulative distance vs gate number



Equidistant fit: distance to t mapping

